

IMAT MOCK EXAM 9: Hard Calibration

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 35/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "Researchers followed two groups of office workers for a year. Group A worked in a building where filtered air was fully exchanged every 20 minutes; Group B worked in an identical building where the air was exchanged every two hours. Temperature, lighting and noise were matched, and neither the workers nor the testers knew which building a participant occupied. Group A scored higher on decision-making tests taken at the end of each working day, with the widest gap on days of high outdoor pollution."

Which statement is most strongly supported by the passage?

- A)** Ventilation rate may influence workers' decision-making, and the difference shows most clearly on days of high outdoor pollution.
- B)** The study proves that outdoor air pollution directly impairs decision-making.
- C)** Temperature, lighting and noise have no effect on decision-making performance.
- D)** The workers in Group A were more motivated than the workers in Group B.
- E)** Indoor air should never be exchanged with outdoor air.

Answer: A

The study controls temperature, lighting and noise and is double-blind, so the performance gap can reasonably be attributed to the ventilation difference, and the passage itself says the gap was widest on high-pollution days. The motivation claim is pure invention, keeping confounders equal does not show they never matter, and a wider gap on polluted days falls well short of proving that outdoor pollution directly impairs decision-making.

2. "Last year a city began charging drivers to enter its central zone. Within six months, car traffic in the zone had fallen by a quarter and buses ran measurably faster. A survey found that most people entering the zone travelled by public transport, and shopkeepers inside the zone reported slightly lower revenues. The money raised from the charge was used to extend two bus routes."

Which statement is best supported by the passage?

- A)** Buses run faster because drivers have abandoned public transport.
- B)** Car journeys into the central zone fell, but the passage does not establish what caused the fall in shop revenues.
- C)** The charge was introduced mainly in order to raise money for bus routes.
- D)** The fall in shop revenues was caused by the traffic charge.
- E)** Most visitors have stopped coming to the central zone.

Answer: B

The passage states the fall in car journeys outright, but it merely reports the shopkeepers' lower revenues alongside it, so no causal link between the charge and the revenue dip is established. The purpose of the charge is never given (only how the revenue was spent), and the survey shows visitors still arrive, mostly by public transport. Reading the revenue fall as caused by the charge confuses a coincidence of facts with a demonstrated cause.

3. Every student who passed the practical examination attended all of the laboratory sessions. Some of the students who attended all of the laboratory sessions were awarded a scholarship. No scholarship winner was charged a laboratory fee.

Which one of the following must be true?

- A) Some students who passed the practical examination were not charged a laboratory fee.
- B) Some scholarship winners passed the practical examination.
- C) All students who were charged a laboratory fee failed the practical examination.
- D) Every student who was not charged a laboratory fee attended all of the laboratory sessions.
- E) Some students who attended all of the laboratory sessions were not charged a laboratory fee.

Answer: E

Givens: every student who passed attended all laboratory sessions; some students who attended all sessions won a scholarship; no scholarship winner was charged a fee.

Deduction 1: the second premise guarantees at least one student who both attended every session and won a scholarship.

Deduction 2: that student, being a scholarship winner, was not charged a fee.

Conclusion: at least one student who attended all of the laboratory sessions was not charged a fee, so that claim is forced.

Why the tempting claim fails: clearing the passers of fees would need some of them to have won scholarships, which no premise guarantees, and declaring every non-charged student an attendee merely reverses the first premise.

4. Exactly one of four office workers jammed the shared printer, and exactly one of the four statements they made about the incident is true. Dana said: Erik jammed the printer. Erik said: Farid did not jam the printer. Farid said: I did not jam the printer. Gus said: Erik did not jam the printer.

Who jammed the printer?

- A) Dana
- B) No assignment fits: the statements are inconsistent.
- C) Erik
- D) Farid
- E) Gus

Answer: D

Givens: exactly one of Dana, Erik, Farid and Gus jammed the printer, and exactly one of the four statements is true.

Deduction 1: if Farid did it, the accusation of Erik is false, Erik's denial about Farid is false, Farid's own denial is false, and Gus's denial about Erik is true, giving exactly one truth.

Deduction 2: if Erik did it, the accusation of Erik, Erik's denial about Farid and Farid's denial are all true, three truths.

Deduction 3: if Dana or Gus did it, then Erik and Farid are both innocent, so the statements of Erik, Farid and Gus are all true, three truths again.

Conclusion: only Farid as the jammer leaves exactly one true statement, so Farid did it.

Why the tempting claim fails: accusing Erik would make Dana's statement true on top of the two denials, breaking the one-truth condition at once.

5. A town installed automatic speed cameras along its main road. During the following year, accidents on that road fell by 30 per cent, and the council concluded that the cameras had slowed drivers down and so prevented accidents.

Which one of the following findings would most weaken the council's argument?

- A) Repairing the cameras cost the council more than it had expected to save.
- B) The cameras recorded several thousand speeding violations during the year.
- C) Accidents on other roads in the same town fell by about 30 per cent over the same year, although no cameras were installed there.
- D) The speed limit on the main road was not changed when the cameras were installed.
- E) Most drivers questioned said that the cameras made them feel constantly watched.

Answer: C

An equally large fall on roads with no cameras points to some town-wide cause (weather, traffic volumes, reporting habits) acting everywhere, which removes the council's grounds for crediting the cameras. Drivers feeling watched and thousands of recorded violations say nothing about what caused the accident drop, and an unchanged speed limit actually closes off an alternative explanation rather than weakening the case.

6. A safety rule states: whenever the fire alarm sounds, all staff evacuate the building.

Which one of the following statements is logically equivalent to this rule?

- A) If the staff evacuate the building, then the fire alarm has sounded.
- B) If the staff do not evacuate the building, then the fire alarm has not sounded.
- C) Either the fire alarm sounds or the staff evacuate the building.
- D) If the fire alarm does not sound, then the staff do not evacuate the building.
- E) The staff evacuate the building if and only if the fire alarm sounds.

Answer: B

Givens: the rule is a one-way conditional: if the alarm sounds, all staff evacuate.

Deduction 1: a conditional of the form $P \text{ implies } Q$ is logically equivalent only to its contrapositive, not- Q implies not- P .

Deduction 2: here that contrapositive reads: if the staff do not evacuate, the fire alarm has not sounded.

Conclusion: the contrapositive is the only logically equivalent statement offered.

Why the tempting claims fail: evacuation implying the alarm sounded runs the arrow backwards, no alarm implying no evacuation only negates both parts without reversing them, the biconditional adds an unproven claim in the reverse direction, and a bare either-or is not a conditional at all.

7. Six friends stand in a row for a photograph, in positions numbered 1 to 6 from left to right. Jana stands in position 1. Femi stands somewhere to the right of Gita. Hugo stands immediately to the left of Kofi. Ivo stands somewhere to the right of Femi.

Which one of the following is NOT possible?

- A) Gita stands in position 5.
- B) Ivo stands in position 4.
- C) Hugo stands in position 2.
- D) Femi stands in position 3.
- E) Kofi stands in position 5.

Answer: A

Givens: Jana fixed in position 1; Gita somewhere left of Femi, Femi somewhere left of Ivo; Hugo immediately left of Kofi; the other five people fill positions 2 to 6.

Deduction 1: the Hugo-Kofi block must occupy positions 2-3, 3-4, 4-5 or 5-6, so there are four cases.

Deduction 2: block at 2-3 forces Gita, Femi, Ivo into positions 4, 5, 6 in that order.

Deduction 3: block at 3-4 forces Gita into 2 and Femi, Ivo into 5, 6.

Deduction 4: block at 4-5 forces Gita, Femi into 2, 3 and Ivo into 6.

Deduction 5: block at 5-6 forces Gita, Femi, Ivo into 2, 3, 4.

Conclusion: these four rows are the only valid arrangements, and Gita lands on position 2 or 4 in every one of them, so Gita in position 5 is impossible.

Why the tempting claims fail: Hugo on 2, Ivo on 4, Femi on 3 and Kofi on 5 each appear in one of the four rows, so none of them is ruled out.

8. The four lectures Anthropology, Botany, Chemistry and Demography are scheduled one per hour at 9:00, 11:00, 13:00 and 15:00. Chemistry is scheduled earlier than Demography. Botany is not scheduled at 9:00. Anthropology is scheduled either immediately before or immediately after Botany.

Which one of the following must be true?

- A) Botany is scheduled at 13:00.
- B) Chemistry is scheduled at 15:00.
- C) Demography is not scheduled at 9:00.
- D) Anthropology is scheduled at 9:00.
- E) Chemistry is scheduled at 11:00.

Answer: C

Givens: lectures at 9:00, 11:00, 13:00 and 15:00; Chemistry earlier than Demography; Botany not at 9:00; Anthropology immediately before or after Botany.

Deduction 1: the Anthropology-Botany pair must occupy adjacent slots, 9:00-11:00, 11:00-13:00 or 13:00-15:00, giving five schedules to test.

Deduction 2: pair at 9:00-11:00 leaves Chemistry at 13:00 and Demography at 15:00.

Deduction 3: pair at 11:00-13:00, either internal order, leaves Chemistry at 9:00 and Demography at 15:00.

Deduction 4: pair at 13:00-15:00, either internal order, leaves Chemistry at 9:00 and Demography at 11:00.

Deduction 5: in all five valid schedules Demography comes after Chemistry, so Demography can never take the first slot.

Conclusion: Demography is not scheduled at 9:00, and this holds in every arrangement, so it must be true.

Why the tempting claims fail: Chemistry sits only at 9:00 or 13:00 (never at 11:00 or 15:00), Botany ranges over 11:00, 13:00 and 15:00, and Anthropology opens the day in only one of the five schedules.

9. Five entries, P, Q, R, S and T, are ranked from 1st (best) to 5th in a short-story competition. T is ranked third. S is ranked better than R. R is ranked better than Q.

Which one of the following must be true?

- A) P is ranked second.
- B) R is ranked fourth.
- C) S is ranked first.
- D) S is ranked better than P.
- E) Q is ranked either fourth or fifth.

Answer: E

Givens: five ranks with T fixed third; the chain S better than R better than Q; P, S, R and Q fill positions 1, 2, 4 and 5.

Deduction 1: Q needs both S and R ahead of it, and position 3 is taken, so Q can sit only fourth or fifth.

Deduction 2: with Q fifth, S and R take two of positions 1, 2 and 4 in that order and P takes the remainder, which gives three rankings.

Deduction 3: with Q fourth, S and R must take positions 1 and 2 and P is pushed to fifth, one ranking.

Deduction 4: in all four resulting rankings Q occupies fourth or fifth place.

Conclusion: Q is ranked either fourth or fifth, and this must be true.

Why the tempting claims fail: one ranking puts P first and S second, so S first, P second and S ahead of P each break, and R holds second place in three of the four rankings, so R fourth fails too.

Section B - Biology (Q10-32)

10. A protein's folded shape is held partly by ionic bonds (salt bridges) between the side chain of a lysine residue and the side chain of an aspartate residue. At pH 7 the lysine side chain is positively charged and the aspartate side chain is negatively charged. The protein is transferred to a solution at pH 2, where the aspartate side chain becomes protonated and loses its charge, while the lysine side chain stays charged. Which outcome is expected?

- A) The primary structure is shortened, because the bond between the two side chains is one kind of peptide bond.
- B) The salt bridge is unaffected, but every peptide bond in the backbone breaks at pH 2.
- C) The salt bridge becomes stronger, because two uncharged groups attract each other more strongly than two charged groups.
- D) The salt bridge breaks and this part of the tertiary structure becomes less stable, while the amino acid sequence is unchanged.
- E) A new salt bridge forms immediately, because at pH 2 the aspartate side chain becomes positively charged.

Answer: D

The salt bridge is an ionic attraction between the positively charged lysine side chain and the negatively charged aspartate side chain. At pH 2 the aspartate side chain is protonated and neutral, so the attraction vanishes and that region of the tertiary structure is destabilised, while the covalent backbone and hence the amino acid sequence are untouched, since side-chain ionic bonds are not peptide bonds. No new bridge can form because aspartate does not turn positive, and uncharged groups do not attract more strongly than opposite charges.

11. A patient takes a drug that keeps the pH of the stomach contents close to 7 instead of the usual strongly acidic value, but leaves the small intestine unchanged. Pepsin works optimally near pH 2, while trypsin works optimally near pH 8. Which outcome is expected?

- A)** Protein digestion continues unchanged, because enzymes adjust their pH optimum to their surroundings.
- B)** Both enzymes lose almost all activity, because digestive enzymes work only in strongly acidic conditions.
- C)** Pepsin activity in the stomach falls sharply, while trypsin activity in the small intestine is essentially unaffected.
- D)** Pepsin activity in the stomach rises, because pepsin neutralises the effect of the drug.
- E)** Trypsin activity in the small intestine falls to zero, because trypsin requires the acidity produced by the stomach.

Answer: C

Pepsin's optimum is near pH 2, so raising the stomach contents to roughly pH 7 leaves it working far below its maximum. The small intestine is unchanged, so trypsin, happiest near pH 8, keeps its normal activity there. Enzymes do not shift their optima to suit their surroundings, and trypsin in any case prefers alkaline conditions, so the claim that both enzymes need strongly acidic conditions is wrong.

12. Glucose units released from glycogen inside a muscle cell enter glycolysis as glucose 6-phosphate, so the hexokinase step, which consumes 1 ATP, is bypassed, while the other ATP-consuming step of glycolysis still occurs. The muscle ferments these units entirely to lactate. What is the net ATP yield per glucose unit?

- A)** 0 **B)** 2 **C)** 3 **D)** 4 **E)** 6

Answer: C

Per glucose unit, glycolysis pays out 4 ATP (2 at phosphoglycerate kinase, 2 at pyruvate kinase). A free glucose molecule would also spend 1 ATP at hexokinase and 1 at phosphofructokinase, but a glycogen-derived unit enters already phosphorylated as glucose 6-phosphate, so only the phosphofructokinase ATP is spent: net = 4 - 1 = 3 ATP. Fermenting pyruvate to lactate merely regenerates NAD⁺ and changes no ATP figure.

13. In the epithelial cells lining the small intestine, glucose is absorbed from the lumen by a transporter that carries glucose together with sodium ions, driven by the sodium concentration gradient. That gradient is maintained by the sodium-potassium pump, which hydrolyses ATP. A poison inactivates the sodium-potassium pump of these cells. Which outcome is expected?

- A)** The cells begin to secrete glucose back into the lumen, because the pump normally carries glucose inward.
- B)** The sodium-potassium pump switches to transporting glucose, so absorption continues unchanged.
- C)** Only the movement of potassium is affected, because sodium and potassium cross the membrane through separate systems.
- D)** The sodium gradient across the membrane decays, so the sodium-coupled glucose transporter stops working and glucose absorption by this route fails.
- E)** Glucose absorption by the sodium-coupled transporter continues at a normal rate, because this transporter does not use ATP directly.

Answer: D

The sodium-glucose cotransporter runs on the downhill sodium gradient, and that gradient exists only because the sodium-potassium pump spends ATP maintaining it (secondary active transport). Inactivate the pump and sodium leaks in until the gradient dissipates, so the cotransporter loses its driving force and glucose absorption by this route fails. The transporter's own lack of direct ATP use is irrelevant: its energy source is the pump.

14. A plant leaf cell is placed in a solution whose concentration of non-penetrating solutes is much higher than that of the cell sap. Which outcome is expected?

- A)** Water enters the cell until the cell bursts, because plant cells have no way of stopping water uptake.
- B)** Water leaves the cell, the protoplast pulls away from the cell wall, and the cell does not burst.
- C)** Water enters the cell and builds turgor pressure against the wall.
- D)** There is no net movement of water, because the cell wall blocks osmosis.
- E)** Water leaves the cell until the cell bursts, because the wall cannot resist shrinking.

Answer: B

The outside solution holds more non-penetrating solute than the cell sap, so it is hypertonic and water leaves the cell by osmosis. The protoplast shrinks and pulls away from the cell wall (plasmolysis), and the cell cannot burst because it is losing water while the rigid wall resists deformation rather than blocking osmosis. Every water-entry picture contradicts the direction of the concentration difference.

15. A drug stops the transcription of ribosomal RNA in the nucleolus of a eukaryotic cell, while the transcription of protein-coding genes and the rest of the cell's metabolism continue intact for several hours. Which outcome is expected?

- A)** Messenger RNA accumulates but is translated less efficiently, because no new ribosomal subunits are assembled, so the rate of protein synthesis falls.
- B)** Messenger RNA is destroyed as soon as it leaves the nucleus, because the nucleolus protects it.
- C)** DNA replication speeds up, because the freed RNA polymerase begins copying DNA.
- D)** Only the synthesis of proteins destined for secretion stops, because those proteins are made by the nucleolus itself.
- E)** Protein synthesis continues unchanged, because ribosomes are assembled in the cytoplasm without the nucleolus.

Answer: A

The nucleolus transcribes ribosomal RNA and assembles ribosomal subunits. With that blocked, no new ribosomes reach the cytoplasm, so messenger RNA keeps being transcribed but is translated ever less efficiently and total protein synthesis falls. Secreted proteins are made on rough ER ribosomes drawing on the same subunit supply, the nucleolus has no protective role for mRNA, and DNA replication is untouched.

16. In a eukaryotic gene, a single base substitution occurs well upstream of the coding sequence, within the promoter region where RNA polymerase binds, and the change weakens this binding. The coding sequence itself is unchanged. Which outcome is expected?

- A)** The reading frame of the gene shifts, so the protein differs completely downstream of the mutation.
- B)** The gene can never be transcribed again, because a damaged promoter destroys the DNA.
- C)** The cell is unaffected in every way, because promoter changes cannot influence protein production.
- D)** One amino acid in the protein is replaced by a different one, because every base substitution changes the protein.
- E)** The amino acid sequence of the protein is unchanged, but the protein is produced in reduced amounts, because transcription of the gene is less efficient.

Answer: E

The substitution lies in the promoter, outside the coding sequence, so the mRNA sequence and therefore the amino acid sequence of the protein are unchanged. Because polymerase binds less well, transcription is less efficient and the protein is made in reduced amounts. Frameshifts need insertions or deletions rather than substitutions, a mutated promoter reduces transcription but does not destroy DNA, and this mutation plainly does affect the cell.

17. During meiosis I in a human oocyte, one pair of homologous chromosomes fails to separate (nondisjunction). Meiosis II then proceeds normally, and the resulting egg cell is fertilised by a normal sperm. Which statement is correct?

- A)** The zygote always has 47 chromosomes, because nondisjunction always produces extra chromosomes.
- B)** The egg is always inviable, because cells with unpaired homologous chromosomes cannot be fertilised.
- C)** The four products have 24, 24, 23 and 23 chromosomes, because only the second division is disturbed.
- D)** Two of the four products are normal, and fertilisation produces a normal zygote in half of the cases.
- E)** All four products of meiosis have an abnormal chromosome number, and fertilisation by a normal sperm produces a zygote with either 45 or 47 chromosomes.

Answer: E

Step 1: in meiosis I nondisjunction both homologues of the pair go to the same pole, so after a normal meiosis II the four products are two with 24 chromosomes and two with 22; none is normal. Step 2: fertilisation by a normal sperm (23 chromosomes) gives 47 or 45 chromosomes depending on which product becomes the egg. The 24, 24, 23, 23 pattern with two normal products belongs to nondisjunction in meiosis II, and aneuploid eggs are certainly capable of being fertilised.

18. Two guinea pigs, both heterozygous for coat colour (Bb) and for hair length (Ll), are crossed. The two genes assort independently. What is the probability that an offspring shows both recessive traits, that is, brown coat (bb) and long hair (ll)?

- A)** 1/16 **B)** 1/8 **C)** 1/4 **D)** 3/16 **E)** 9/16

Answer: A

For each gene separately, Bb × Bb gives the recessive phenotype with probability 1/4.

The genes assort independently, so $P(bb \text{ and } ll) = 1/4 \times 1/4 = 1/16$

The value 9/16 is instead the chance of showing both dominant traits.

19. In a population in Hardy-Weinberg equilibrium, one child in 2,500 is born with a recessive genetic disease. Assuming random mating and a large population, roughly what fraction of the population are carriers (heterozygotes) for the disease allele?

- A)** 1/2500 **B)** 1/50 **C)** 1/25 **D)** 2/25 **E)** 49/50

Answer: C

Affected frequency $q^2 = 1/2500$, so $q = 1/50 = 0.02$ and $p = 0.98$

Carrier frequency = $2pq = 2 \times 0.98 \times 0.02 = 0.0392$, which is roughly $0.04 = 1/25$

The allele frequency itself is 1/50, and 1/2500 counts only affected individuals, so both undershoot the heterozygote total.

20. Colchicine prevents the assembly of spindle microtubules. A cell that has completed DNA replication is exposed to colchicine as it enters mitosis. At which stage, and with which appearance, will treated cells accumulate?

- A)** In anaphase, because colchicine triggers the separation of the chromatids.
- B)** In interphase, because the cell returns to G1 without dividing.
- C)** In prophase, because spindle formation is required for chromosome condensation.
- D)** At metaphase, with fully condensed chromosomes each consisting of two sister chromatids and no separation of the chromatids.
- E)** In telophase, because the nuclear envelope reforms without spindle function.

Answer: D

Chromosomes condense and the nuclear envelope breaks down without needing the spindle, so treated cells reach a metaphase-like state with fully condensed chromosomes, each still two joined sister chromatids. With no microtubules nothing pulls the chromatids apart, anaphase never begins and the cells pile up there; colchicine blocks division rather than triggering it or sending cells back to interphase.

21. A toxin blocks the proton channel of mitochondrial ATP synthase, so protons can no longer flow back into the matrix, while the electron transport chain still transfers electrons to oxygen at first. In cells whose mitochondria are treated with this toxin, which statement is correct?

- A)** The cells switch to photosynthesis in order to regenerate their NADPH.
- B)** Electron transport soon stops because the growing proton gradient opposes further pumping, NADH accumulates and is no longer reoxidised, and glycolysis can only continue by converting pyruvate to lactate.
- C)** Glycolysis continues normally, because glycolysis never needs NAD⁺ at any step.
- D)** The proton gradient dissipates through the blocked ATP synthase and electron transport accelerates above normal.
- E)** Electron transport continues at its normal rate, because oxygen is still available and ATP synthesis is a separate process.

Answer: B

Step 1: with the proton return path blocked, the gradient builds until protons can no longer be pumped, so electron transport stalls even though oxygen is present. Step 2: NADH is no longer reoxidised and accumulates. Step 3: glycolysis needs NAD⁺ at its oxidation step, so the cell regenerates NAD⁺ by reducing pyruvate to lactate. The rival story, a dissipating gradient with accelerated transport, describes an uncoupler, which is the opposite of a blocked channel.

22. After death, calcium ions leak out of the sarcoplasmic reticulum and are never pumped back, and ATP production has stopped. Cross-bridges form between myosin heads and actin. Which statement explains why the muscles then become rigid (rigor mortis)?

- A)** The actin filaments shorten permanently, pulling the muscle rigid.
- B)** Without ATP the myosin heads cannot bind actin at all, so the fibres separate and harden.
- C)** The lack of ATP causes the sarcoplasmic reticulum to release more calcium, which stiffens the muscle membrane.
- D)** Tropomyosin detaches from actin because its position depends on ATP rather than on calcium.
- E)** Myosin heads remain locked onto actin because ATP is needed to detach them, and with no ATP the cross-bridges cannot break, so the muscle stays contracted.

Answer: E

In the cross-bridge cycle, ATP binding to myosin is what detaches the head from actin. After death there is no ATP, so heads that have bound actin can never release: the cross-bridges lock and the muscle freezes in a contracted, rigid state. Filament lengths never change, the calcium has already leaked out of the sarcoplasmic reticulum, and tropomyosin's position depends on calcium, not ATP.

23. The aortic semilunar valve of a patient is narrowed (stenosed), so the opening through which blood leaves the left ventricle is reduced. Which chain of consequences follows?

- A)** Blood flows back from the aorta into the left ventricle during diastole, and the right ventricle enlarges first.
- B)** The left ventricle must generate a higher pressure to eject blood, ejection takes longer, and the ventricle wall thickens over time under the extra load.
- C)** The left atrium enlarges because it pumps blood directly into the aorta.
- D)** The mitral valve is held open during systole, which raises the pressure in the left atrium.
- E)** Blood pressure in the pulmonary arteries falls, because the lungs receive less blood.

Answer: B

A narrowed outlet raises the resistance the left ventricle works against, so it must reach a higher pressure to eject, ejection is prolonged, and the wall thickens over time under the chronic load. Backflow from the aorta during diastole describes an incompetent (leaky) valve, not a narrowed one; the mitral valve is closed throughout systole; and the pulmonary circuit is not the first circuit to feel the problem.

24. In a healthy chest, the pressure in the pleural cavity between the lung and the chest wall is lower than atmospheric, which keeps the lung expanded against the chest wall. A fractured rib tears the lung surface and lets air leak into the right pleural cavity until the pressure there equals atmospheric pressure. What happens to the right lung, and why?

- A)** The right lung collapses inward, because its elastic recoil is no longer opposed by the sub-atmospheric pleural pressure, while the chest wall springs outward.
- B)** The lung and the chest wall are drawn toward each other with extra force, keeping the lung inflated.
- C)** The right lung inflates further, because the incoming air pushes directly into the lung tissue.
- D)** The right lung collapses because the air destroys the elastic fibres of the alveolar walls.
- E)** The right lung is unaffected, because the diaphragm alone keeps it inflated.

Answer: A

Sub-atmospheric pleural pressure normally couples the lung's inward elastic recoil to the chest wall, keeping the lung expanded. When air leaks in and the pressure equals atmospheric, the recoil is unopposed, so the lung collapses inward while the chest wall springs outward. Nothing pushes air into the lung tissue and the alveolar elastic fibres are not destroyed; with no pressure difference there is simply nothing holding the lung open.

25. Carbon monoxide binds to haemoglobin at the same sites as oxygen, with an affinity more than 200 times greater, and it is released only slowly. A fire victim is breathing air containing carbon monoxide. A pulse oximeter cannot distinguish carboxyhaemoglobin from oxyhaemoglobin. Which statement is correct?

- A)** Tissue oxygen delivery rises, because carbon monoxide increases the affinity of haemoglobin for oxygen.
- B)** The blood carries less oxygen, and a pulse oximeter immediately shows a falsely low reading, which makes diagnosis simple.
- C)** Oxygen transport is normal, because carbon monoxide displaces carbon dioxide from haemoglobin rather than oxygen.
- D)** The blood carries far less oxygen than its haemoglobin capacity allows, yet a pulse oximeter still reads a high saturation because carbon monoxide bound to haemoglobin looks like oxygen-bound haemoglobin.
- E)** Carbon monoxide only reduces the small amount of oxygen dissolved in plasma, leaving haemoglobin transport untouched.

Answer: D

Carbon monoxide sits on the same haem sites as oxygen with vastly higher affinity, so most haemoglobin is locked up as carboxyhaemoglobin and the blood's oxygen content collapses. A pulse oximeter cannot optically distinguish carboxyhaemoglobin from oxyhaemoglobin, so it reports a falsely reassuring high saturation. Dissolved plasma oxygen is a negligible share of transport, and the remaining sites hold oxygen more tightly, releasing less to tissues, so delivery falls rather than rises.

26. A glucose molecule is absorbed from the lumen of the small intestine into a capillary of an intestinal villus. Which of the following gives the correct route it takes on the way to the hepatic vein?

- A)** Intestinal villus capillary, hepatic artery, liver sinusoids, hepatic portal vein
- B)** Intestinal villus capillary, hepatic vein, liver sinusoids, hepatic portal vein
- C)** Intestinal villus capillary, hepatic portal vein, liver sinusoids, hepatic vein
- D)** Intestinal villus capillary, vena cava, right atrium, hepatic portal vein
- E)** Intestinal villus capillary, lacteal, thoracic duct, hepatic vein

Answer: C

Glucose entering a blood capillary of the villus travels in the hepatic portal vein to the liver, crosses the liver sinusoids (a second capillary bed) and leaves by the hepatic vein, without passing the heart on the way. The lacteal carries fats in lymph rather than glucose, and the hepatic artery brings oxygenated blood to the liver instead of collecting absorbed nutrients.

27. A person takes tablets containing thyroxine (T4) every day in fairly high doses, without medical supervision. The normal feedback control of the thyroid axis still works correctly. Which pattern of blood hormone levels is expected?

- A)** TSH low and T4 high, because the extra T4 suppresses the hypothalamus and the pituitary through normal negative feedback.
- B)** TSH and T4 both normal, because thyroxine taken by mouth is destroyed before reaching the blood.
- C)** TSH high and T4 high, because the pituitary reacts to high T4 by secreting more TSH.
- D)** TRH and TSH both high, because the thyroid gland cannot respond to the tablets.
- E)** TSH high and T4 low, because the tablets stop the thyroid gland from working.

Answer: A

The tablets raise circulating T4 above normal. Because feedback control is intact, the hypothalamus and pituitary respond to high T4 by cutting TRH and TSH secretion, so TSH is low while T4 is high. The patterns with high TSH get the feedback backwards, and oral thyroxine is absorbed (it is exactly how hypothyroidism is treated), so the blood levels are anything but normal.

28. A hill walker walks for hours in extreme heat, loses a large volume of sweat and drinks almost nothing. Plasma volume falls and the plasma becomes more concentrated. Which set of responses is expected?

- A)** Aldosterone secretion falls, so more sodium is lost in the urine.
- B)** The glomerular filtration rate rises sharply and urine volume increases.
- C)** ADH secretion stays unchanged, because ADH responds only to blood pressure and never to plasma concentration.
- D)** ADH secretion falls and the kidneys produce large volumes of dilute urine.
- E)** ADH secretion rises, the collecting ducts reabsorb more water so the urine becomes concentrated, and thirst is stimulated.

Answer: E

Sweating without drinking makes the plasma more concentrated and reduces its volume, and both changes stimulate ADH release from the posterior pituitary. ADH makes the collecting ducts reabsorb water, so the urine becomes scant and concentrated, and hypothalamic osmoreceptors produce thirst. Falling ADH with copious dilute urine describes the response to drinking too much, the opposite scenario.

29. A person with type 1 diabetes injects insulin but then skips the next meal and takes vigorous exercise, and the blood glucose level begins to fall below normal. Which sequence of responses follows in this person?

- A)** The pancreas secretes additional insulin, which corrects the glucose level.
- B)** Glucagon and adrenaline secretion rise, the liver breaks down glycogen and releases glucose into the blood, and the person feels warning symptoms such as hunger and trembling.
- C)** The liver converts glucose into glycogen faster, storing away the effect of the injected insulin.
- D)** The kidneys stop filtering glucose, which keeps the blood level constant.
- E)** Glucagon secretion falls, so the muscles take up more glucose from the blood.

Answer: B

Falling blood glucose triggers the counter-regulatory hormones glucagon and adrenaline, which mobilise liver glycogen into glucose and produce the warning symptoms of hunger, trembling and sweating. In type 1 diabetes the pancreas cannot secrete extra insulin, storing glucose as glycogen would push the level the wrong way, and falling glucagon would deepen the hypoglycaemia.

30. The oesophagus is lined by stratified squamous epithelium many cells thick, while the small intestine is lined by a single layer of columnar cells. Which statement best explains this difference?

- A)** The two linings do the same job, and the difference is a random outcome of embryonic development.
- B)** A single layer of columnar cells is stronger, which suits the intestine as the first part of the gut to receive food.
- C)** Stratified squamous epithelium secretes more digestive enzymes, which is why the oesophagus digests food first.
- D)** The oesophagus needs a many-layered, abrasion-resistant lining against swallowed food, while the intestine needs a thin single layer specialised for absorption.
- E)** The oesophagus absorbs nutrients more slowly than the intestine and needs extra layers to compensate.

Answer: D

Structure follows function: the oesophagus is scraped by swallowed food, so a many-layered squamous lining sacrifices absorption for abrasion resistance, while the intestine absorbs nutrients through a single thin layer that keeps diffusion distances short. The oesophagus digests nothing, and the single layer is a functional specialisation rather than a weakness.

31. A patient tears the meniscus, a pad of cartilage in the knee, and is told that it heals much more slowly than a similar tear in the skin. Which feature of cartilage best explains this?

- A)** Cartilage is buried under skin everywhere in the body, so repair cells cannot physically reach it.
- B)** The cartilage matrix is made of keratin, which never grows back after damage.
- C)** Cartilage cells divide faster than skin cells but build a weaker matrix.
- D)** Cartilage contains few or no blood vessels, so repair cells and their building materials reach the injury only slowly by diffusion.
- E)** Cartilage is a type of nervous tissue, so repair signals cannot reach the injury.

Answer: D

Repair cells, oxygen and matrix building blocks reach a wound through blood vessels, and cartilage is essentially avascular, so supplies arrive only by slow diffusion through the matrix and healing drags on. Keratin is an epidermal protein, cartilage is connective tissue, and sitting beneath the skin does not physically block repair cells from reaching it.

32. A large beetle population has an allele frequency of 0.4 for allele A at one locus. A flood leaves only ten survivors: six AA, three Aa and one aa, and these found a new population. What is the frequency of allele A among the survivors, and what does the change from 0.4 illustrate?

- A)** 0.15, and the change illustrates gene flow from another population.
- B)** 0.40, and the change illustrates the constancy required by the Hardy-Weinberg principle.
- C)** 0.60, and the change illustrates natural selection favouring allele A.
- D)** 0.75, and the change illustrates genetic drift caused by a population bottleneck.
- E)** 0.80, and the change illustrates a new mutation increasing allele A.

Answer: D

The ten survivors hold $6 \times 2 + 3 = 15$ copies of allele A out of 20 alleles, a frequency of $15/20 = 0.75$. A chance-driven crash to a handful of founders that shifts allele frequency is the bottleneck form of genetic drift. Counting only the homozygotes' alleles gives $12/20 = 0.60$, and nothing in the story involves migration, selection or mutation.

Section C - Chemistry (Q33-47)

33. Consider three pairs of organic compounds. Pair P: butan-1-ol and butan-2-ol. Pair Q: (2R)-butan-2-ol and (2S)-butan-2-ol. Pair R: cis-pent-2-ene and trans-pent-2-ene. Which of the following statements is correct?

- A)** P and Q are pairs of enantiomers, while R is a pair of constitutional isomers.
- B)** All three pairs are constitutional isomers, since each pair shares the same molecular formula.
- C)** R is a pair of enantiomers, because the cis and trans forms are mirror images of each other.
- D)** P is a pair of stereoisomers, since the two alcohols differ only in the position of the OH group.
- E)** Q and R are pairs of stereoisomers, while P is a pair of constitutional isomers.

Answer: E

Butan-1-ol and butan-2-ol share a formula but differ in which carbon bears the OH group, so they are constitutional isomers. The R and S forms of butan-2-ol are non-superimposable mirror images (enantiomers), and the cis and trans forms of pent-2-ene differ in the geometry about the double bond, so both of those pairs are stereoisomers. Cis and trans alkenes are diastereomers rather than mirror images, and a difference in position of a group never produces stereoisomers.

34. 4.8 g of methane ($M = 16 \text{ g/mol}$) is mixed with 4.8 g of oxygen gas ($M = 32 \text{ g/mol}$) and ignited, and the reaction $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ proceeds until one reactant is used up. Which statement is correct? (Molar volume of any gas at STP = $22.4 \text{ dm}^3/\text{mol}$)

- A)** 1.68 dm^3 of CO_2 (measured at STP) forms and 3.6 g of methane remains.
- B)** 1.68 dm^3 of CO_2 (measured at STP) forms and no methane remains.
- C)** 1.68 dm^3 of CO_2 (measured at STP) forms and 1.8 g of methane remains.
- D)** 3.36 dm^3 of CO_2 (measured at STP) forms and the oxygen is in excess.
- E)** 3.36 dm^3 of CO_2 (measured at STP) forms and 1.8 g of methane remains.

Answer: A

Given: 4.8 g of CH_4 ($M = 16 \text{ g/mol}$), 4.8 g of O_2 ($M = 32 \text{ g/mol}$), equation $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, molar volume $22.4 \text{ dm}^3/\text{mol}$ at STP.

Relationship: $n = m \div M$; the equation fixes a 1 : 2 : 1 ratio of methane to oxygen to carbon dioxide; gas volume = $n \times 22.4 \text{ dm}^3$ at STP.

Substitute: $n(\text{CH}_4) = 4.8 \div 16 = 0.30 \text{ mol}$; $n(\text{O}_2) = 4.8 \div 32 = 0.15 \text{ mol}$

Limiting check: 0.15 mol of O_2 burns only $0.15 \div 2 = 0.075 \text{ mol}$ of CH_4 , so the oxygen runs out first.

Hop: CO_2 formed = 0.075 mol (1:1 with the methane burned), so $V = 0.075 \times 22.4 = 1.68 \text{ dm}^3$ at STP.

Hop: methane left = $0.30 - 0.075 = 0.225 \text{ mol}$, mass = $0.225 \times 16 = 3.6 \text{ g}$

Result: 1.68 dm^3 of CO_2 forms at STP and 3.6 g of methane remains.

Classic slip: letting the 0.15 mol of oxygen stand for 0.15 mol of CO_2 , which inflates the volume to 3.36 dm^3 by ignoring the 2:1 oxygen requirement.

35. 0.40 g of sodium hydroxide ($M = 40 \text{ g/mol}$), a strong base that is fully dissociated in water, is dissolved and made up to exactly 1.0 dm^3 of solution at 25°C . Then 10 cm^3 of this solution is diluted with water to a final volume of 1.0 dm^3 . What is the pH of the final solution? ($K_w = 1.0 \times 10^{-14}$ at 25°C)

- A)** 8 **B)** 9 **C)** 10 **D)** 11 **E)** 12

Answer: C

Given: 0.40 g of NaOH ($M = 40 \text{ g/mol}$) made up to 1.0 dm^3 , then 10 cm^3 of it diluted to 1.0 dm^3 ; $K_w = 1.0 \times 10^{-14}$ at 25°C .

Relationship: $n = m \div M$; NaOH is fully dissociated so $[\text{OH}^-]$ equals the NaOH concentration; $\text{pH} + \text{pOH} = 14$; dilution divides the concentration by the volume factor.

Substitute: $n = 0.40 \div 40 = 0.010 \text{ mol}$, so the first solution has $[\text{OH}^-] = 0.010 \div 1.0 = 1.0 \times 10^{-2} \text{ mol/dm}^3$

Hop: $\text{pOH} = 2$, so $\text{pH} = 14 - 2 = 12$ in the first solution

Dilution: factor = $1000 \div 10 = 100$, so $[\text{OH}^-] = 1.0 \times 10^{-2} \div 100 = 1.0 \times 10^{-4} \text{ mol/dm}^3$

Hop: $\text{pOH} = 4$

Result: $\text{pH} = 14 - 4 = 10$

Classic slip: reading the final pOH , 4, as the pH , or treating the hundredfold dilution as tenfold, which would give pH 11.

36. At 25 °C the ionic product of water is $K_w = 1.0 \times 10^{-14}$, but at a higher temperature K_w is larger: in one such case $K_w = 1.0 \times 10^{-13}$. What is the pH of a 0.010 mol/dm³ solution of sodium hydroxide, fully dissociated, at this higher temperature?

- A) 2 B) 6.5 C) 11 D) 12 E) 13**

Answer: C

Given: $K_w = 1.0 \times 10^{-13}$ at the higher temperature; $[\text{OH}^-] = 0.010 \text{ mol/dm}^3$ from fully dissociated NaOH.

Relationship: $K_w = [\text{H}^+] \times [\text{OH}^-]$, so $[\text{H}^+] = K_w \div [\text{OH}^-]$, and $\text{pH} = -\log_{10}[\text{H}^+]$

Substitute: $[\text{H}^+] = (1.0 \times 10^{-13}) \div (1.0 \times 10^{-2})$

Hop: $[\text{H}^+] = 1.0 \times 10^{-11} \text{ mol/dm}^3$

Hop: $\text{pH} = -\log_{10}(1.0 \times 10^{-11}) = 11$

Result: $\text{pH} = 11$ at this temperature

Classic slip: dividing by the 25 °C value of K_w , 1.0×10^{-14} , which lands on pH 12; the solution is still firmly alkaline even though neutral water here would read pH 6.5.

37. 20 cm³ of 0.010 mol/dm³ hydrochloric acid, a strong acid that is fully ionised, is diluted with water to a final volume of 400 cm³. Taking $\log_{10} 2 = 0.30$, what is the pH of the diluted solution?

- A) 2.30 B) 3.00 C) 3.10 D) 3.30 E) 4.00**

Answer: D

Given: 20 cm³ of 0.010 mol/dm³ HCl (strong, fully ionised) diluted to 400 cm³; $\log_{10} 2 = 0.30$.

Relationship: the dilution factor is $V(\text{final}) \div V(\text{initial})$; $[\text{H}^+]$ is divided by it, and each tenfold drop in $[\text{H}^+]$ raises pH by exactly 1, so $\Delta\text{pH} = \log_{10}(\text{dilution factor})$.

Substitute: dilution factor = $400 \div 20 = 20$

Hop: initial $[\text{H}^+] = 0.010 \text{ mol/dm}^3$, so initial $\text{pH} = 2.00$

Hop: $\Delta\text{pH} = \log_{10} 20 = \log_{10} 2 + \log_{10} 10 = 0.30 + 1.00 = 1.30$

Result: $\text{pH} = 2.00 + 1.30 = 3.30$ (check: $[\text{H}^+] = 0.010 \div 20 = 5.0 \times 10^{-4} \text{ mol/dm}^3$, and $-\log(5.0 \times 10^{-4}) = 3.30$)

Classic slip: assuming a clean tenfold dilution, which gives pH 3.00, or a hundredfold one, which gives 4.00; the true factor of 20 needs the log of 2.

38. Four pure molecular liquids with similar molar masses are: propane $\text{CH}_3\text{CH}_2\text{CH}_3$ (Mr 44), propanone CH_3COCH_3 (Mr 58), ethanol $\text{C}_2\text{H}_5\text{OH}$ (Mr 46) and ethanoic acid CH_3COOH (Mr 60). In the liquid state, two ethanoic acid molecules are held together by two hydrogen bonds each, and propanone has no O-H or N-H bond. Which option places them in order of increasing boiling point?

- A) ethanoic acid < ethanol < propanone < propane**
B) propane < propanone < ethanol < ethanoic acid
C) propane < propanone < ethanoic acid < ethanol
D) propanone < propane < ethanol < ethanoic acid
E) propane < ethanol < propanone < ethanoic acid

Answer: B

Boiling point tracks the strongest intermolecular force. Propane is non-polar (London forces only) and boils lowest; propanone adds a permanent dipole but has no O-H bond, so it cannot donate hydrogen bonds; ethanol hydrogen bonds in pairs; and ethanoic acid forms dimers held by two hydrogen bonds at once, needing the most energy to vaporise. The order is propane, then propanone, then ethanol, then ethanoic acid; the similar molar masses mean size cannot decide the ranking.

39. For the reaction $A + 2B \rightarrow \text{products}$, experiments at fixed temperature show that the rate is proportional to the concentration of A and to the square of the concentration of B. If the concentration of A is doubled and the concentration of B is tripled, by what factor does the reaction rate increase?

- A) 6 B) 8 C) 9 D) 12 E) 18**

Answer: E

Given: rate proportional to $[A]^1$ and to $[B]^2$; [A] doubled; [B] tripled.

Relationship: total rate factor = (change in A)^(order in A) $\times$ (change in B)^(order in B) = $2^1 \times 3^2$.

Hop: $3^2 = 9$

Hop: total = $2 \times 9 = 18$

Result: the rate increases by a factor of 18.

Classic slip: dropping the square and computing $2 \times 3 = 6$, which treats the rate as first order in B.

40. What is the average oxidation number of a sulfur atom in the tetrathionate ion, $S_4O_6^{2-}$, if the four sulfur atoms are treated as equivalent?

- A) +2 B) +2.5 C) +3 D) +5 E) +10**

Answer: B

Given: tetrathionate ion $S_4O_6^{2-}$, four sulfur atoms treated as equivalent, oxygen = -2.

Relationship: oxidation numbers sum to the overall charge: $4x + 6 \times (-2) = -2$.

Substitute: $4x - 12 = -2$

Hop: $4x = -2 + 12 = 10$

Result: $x = 10 \div 4 = +2.5$, the average oxidation number per sulfur atom.

Classic slip: reporting +10, the total held by all four sulfur atoms together, as if it were one atom's average.

41. In which species does nitrogen have the same oxidation number as sulfur has in the sulfite ion, SO_3^{2-} ?

- A) NO₂ B) NH₃ C) NO D) N₂ E) N₂O**

Answer: A

Given: sulfur in SO_3^{2-} ; nitrogen in N_2 , NH_3 , NO , NO_2 and N_2O ; oxygen = -2; neutral molecules sum to 0.

Relationship: the sum of oxidation numbers equals the species' overall charge.

Hop 1: sulfur, $x + 3 \times (-2) = -2$, so $x = +4$

Hop 2: nitrogen in NO_2 , $x + 2 \times (-2) = 0$, so $x = +4$: a match

Hop 3: sweep the rest: N_2 is 0 (element), NH_3 gives $x + 3 = 0$ so $x = -3$, NO gives $x - 2 = 0$ so $x = +2$, N_2O gives $2x - 2 = 0$ so $x = +1$.

Result: only in NO_2 does nitrogen carry +4, the same as sulfur in the sulfite ion.

Classic slip: forgetting the -2 charge when computing sulfur, which gives +6 instead of +4 and sends the matching astray.

42. A substance decomposes by first-order kinetics with a half-life of 20 minutes at 25 °C. A sealed 2.0 dm³ vessel initially contains 0.80 mol of the substance. What is its concentration after 60 minutes?

- A)** 0.025 mol/dm³ **B)** 0.040 mol/dm³ **C)** 0.050 mol/dm³ **D)** 0.10 mol/dm³ **E)** 0.20 mol/dm³

Answer: C

Given: first-order kinetics, half-life 20 minutes, elapsed time 60 minutes, initial amount 0.80 mol in a 2.0 dm³ vessel.

Relationship: each half-life halves the amount; number of half-lives = time ÷ half-life; concentration = amount ÷ volume.

Substitute: number of half-lives = $60 \div 20 = 3$

Hop 1: after 20 min, $0.80 \div 2 = 0.40$ mol

Hop 2: after 40 min, $0.40 \div 2 = 0.20$ mol

Hop 3: after 60 min, $0.20 \div 2 = 0.10$ mol

Hop 4: concentration = $0.10 \div 2.0 = 0.050$ mol/dm³

Result: 0.050 mol/dm³ after 60 minutes.

Classic slip: quoting the 0.10 as a concentration without dividing by the 2.0 dm³ vessel, or halving one time more, which would give 0.025 mol/dm³.

43. Sample X contains 4.4 g of carbon dioxide, CO₂, and sample Y contains 4.9 g of sulfuric acid, H₂SO₄ (M = 98 g/mol). (C = 12, O = 16; Avogadro constant = 6.02×10^{23} per mol.) Which statement about the oxygen atoms in the two samples is correct?

- A)** Each sample contains 1.2×10^{23} oxygen atoms.
B) Each sample contains 2.4×10^{23} oxygen atoms.
C) Sample X contains 2.4×10^{23} oxygen atoms and sample Y contains 1.2×10^{23} .
D) Each sample contains 6.0×10^{22} oxygen atoms.
E) Sample X contains 1.2×10^{23} oxygen atoms and sample Y contains 2.4×10^{23} .

Answer: A

Given: X = 4.4 g of CO₂ (M = 12 + 32 = 44 g/mol); Y = 4.9 g of H₂SO₄ (M = 98 g/mol); Avogadro constant 6.02×10^{23} per mol

Relationship: $n = m \div M$; each CO₂ holds 2 oxygen atoms, each H₂SO₄ holds 4; atoms = $n \times \text{atoms per molecule} \times 6.02 \times 10^{23}$.

Substitute X: $n = 4.4 \div 44 = 0.10$ mol of CO₂, giving $0.10 \times 2 = 0.20$ mol of oxygen atoms

Substitute Y: $n = 4.9 \div 98 = 0.050$ mol of H₂SO₄, giving $0.050 \times 4 = 0.20$ mol of oxygen atoms

Hop: $0.20 \times 6.02 \times 10^{23} = 1.204 \times 10^{23} \approx 1.2 \times 10^{23}$ atoms in each sample

Result: both samples contain about 1.2×10^{23} oxygen atoms.

Classic slip: doubling one conversion, for instance crediting each CO₂ with 4 oxygens, which yields 2.4×10^{23} for one sample and breaks the equality.

44. 50 cm³ of 0.20 mol/dm³ hydrochloric acid, fully ionised, is mixed with 150 cm³ of 0.10 mol/dm³ sodium hydroxide, fully dissociated. What is the pH of the resulting solution at 25 °C? ($K_w = 1.0 \times 10^{-14}$; $\log 2.5 = 0.40$)

- A)** 1.60 **B)** 11.60 **C)** 12.00 **D)** 12.40 **E)** 13.00

Answer: D

Given: 50 cm³ of 0.20 mol/dm³ HCl; 150 cm³ of 0.10 mol/dm³ NaOH; both fully ionised; $K_w = 1.0 \times 10^{-14}$; $\log 2.5 = 0.40$.

Relationship: $n = c \times V$; H^+ and OH^- neutralise 1:1; the surviving ion's concentration = leftover moles $\div$ total volume; $pH + pOH = 14$.

Substitute: $n(H^+) = 0.050 \times 0.20 = 0.010$ mol; $n(OH^-) = 0.150 \times 0.10 = 0.015$ mol

Hop: excess $OH^- = 0.015 - 0.010 = 0.005$ mol (the base wins)

Hop: total volume = 50 + 150 = 200 cm³ = 0.200 dm³, so $[OH^-] = 0.005 \div 0.200 = 0.025$ mol/dm³

Hop: $pOH = -\log(2.5 \times 10^{-2}) = 2 - 0.40 = 1.60$

Result: $pH = 14 - 1.60 = 12.40$

Classic slip: skipping the division by the combined volume, which inflates $[OH^-]$ to 0.005 mol/dm³ and the pH to 13.00, or reporting the pOH of 1.60 as the pH.

45. 1.08 g of aluminium ($M = 27$ g/mol) is added to 100 cm³ of 0.10 mol/dm³ sulfuric acid and reacts completely as far as possible: $2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$. Which statement is correct? (Molar volume of a gas at STP = 22.4 dm³/mol)

- A)** 0.224 dm³ of H₂ (measured at STP) is released and the acid is in excess.
B) 0.224 dm³ of H₂ (measured at STP) is released and aluminium is in excess.
C) 0.224 dm³ of H₂ (measured at STP) is released and no aluminium remains.
D) 0.448 dm³ of H₂ (measured at STP) is released and aluminium is in excess.
E) 1.344 dm³ of H₂ (measured at STP) is released and the acid is in excess.

Answer: B

Given: 1.08 g of Al ($M = 27$ g/mol); 100 cm³ of 0.10 mol/dm³ H₂SO₄; equation $2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$; molar volume 22.4 dm³/mol at STP.

Relationship: $n = m \div M$ or $c \times V$; the reagent that runs out first fixes the yield; H₂ forms 1:1 with the acid consumed (3:3 in the equation).

Substitute: $n(Al) = 1.08 \div 27 = 0.040$ mol; $n(H_2SO_4) = 0.100 \times 0.10 = 0.010$ mol

Limiting check: consuming all the aluminium would need $0.040 \times 3 \div 2 = 0.060$ mol of acid, but only 0.010 mol exists, so the acid is limiting.

Hop: H₂ formed = 0.010 mol (1:1 with the acid)

Hop: $V = 0.010 \times 22.4 = 0.224$ dm³ at STP

Hop: aluminium consumed = $0.010 \times 2 \div 3 \approx 0.0067$ mol $\approx$ 0.18 g, leaving about $1.08 - 0.18 = 0.90$ g of aluminium unused.

Result: 0.224 dm³ of H₂ at STP, with aluminium in excess.

Classic slip: treating the 0.040 mol of aluminium as limiting, which multiplies the gas up to 1.344 dm³ and ignores the acid running dry.

46. A student evaporates water from 500 cm³ of a 0.20 mol/dm³ sodium chloride solution until the volume is exactly 200 cm³, with no loss of solute. The concentration of the final solution is:

- A)** 0.08 mol/dm³ **B)** 0.10 mol/dm³ **C)** 0.20 mol/dm³ **D)** 0.40 mol/dm³ **E)** 0.50 mol/dm³

Answer: E

Given: 500 cm³ of 0.20 mol/dm³ NaCl evaporated to exactly 200 cm³, no solute lost.

Relationship: $n = c \times V$; evaporation removes solvent only, so n stays fixed and $c(\text{new}) = n \div V(\text{new})$.

Substitute: $n = 0.500 \text{ dm}^3 \times 0.20 \text{ mol/dm}^3 = 0.10 \text{ mol}$

Hop: $V(\text{new}) = 200 \text{ cm}^3 = 0.200 \text{ dm}^3$

Result: $c(\text{new}) = 0.10 \div 0.200 = 0.50 \text{ mol/dm}^3$

Classic slip: multiplying by the volume ratio instead of dividing ($0.20 \times 200/500 = 0.08 \text{ mol/dm}^3$), which reverses what removing water does.

47. Two solutions at 25 °C have pH values of 3.5 and 5.5. The hydrogen ion concentration of the solution with pH 3.5 is how many times that of the solution with pH 5.5?

- A)** 2 **B)** 20 **C)** 100 **D)** 1000 **E)** 10000

Answer: C

Given: two solutions at pH 3.5 and pH 5.5 at 25 °C.

Relationship: $\text{pH} = -\log_{10}[\text{H}^+]$, so each whole unit of pH is a factor of 10 in $[\text{H}^+]$, and the factor between two solutions is 10 raised to the pH difference.

Substitute: difference = $5.5 - 3.5 = 2.0$ units

Hop: factor = $10^2 = 100$

Result: the pH 3.5 solution has 100 times the hydrogen ion concentration of the pH 5.5 solution.

Classic slip: treating the pH scale as linear and multiplying the 2-unit gap by ten to get 20.

Section D - Mathematics (Q48-54)

48. Given that $\log_{10} 5 = 0.70$ and $\log_{10} 3 = 0.48$, what is the value of $\log_{10} 45$?

- A)** 1.18 **B)** 1.30 **C)** 1.66 **D)** 1.88 **E)** 2.18

Answer: C

Given: $\log_{10} 5 = 0.70$; $\log_{10} 3 = 0.48$

Relationship: $45 = 3^2 \times 5$, so $\log_{10} 45 = 2 \times \log_{10} 3 + \log_{10} 5$ (product rule plus power rule).

Substitute: $\log_{10} 45 = 2 \times 0.48 + 0.70$

Hop: $2 \times 0.48 = 0.96$

Result: $0.96 + 0.70 = 1.66$

Classic slip: reading 45 as 3×5^2 instead, which gives $0.48 + 2 \times 0.70 = 1.88$, or as 3×5 , which gives 1.18.

49. For which values of k does the equation $x^2 + kx + 9 = 0$ have no real solutions?

- A) $k \leq -6$ or $k \geq 6$
- B) $-6 \leq k \leq 6$
- C) all real values of k
- D) $k < -6$ or $k > 6$
- E) $-6 < k < 6$

Answer: E

Given: $x^2 + kx + 9 = 0$, asking for the values of k that give no real solutions.

Relationship: a quadratic has no real roots exactly when its discriminant $k^2 - 4ac$ is negative.

Substitute: discriminant $= k^2 - 4 \times 1 \times 9 = k^2 - 36$

Condition: $k^2 - 36 < 0$, that is $k^2 < 36$.

Hop: $|k| < 6$, so $-6 < k < 6$ with both ends excluded.

Result: no real solutions precisely when $-6 < k < 6$.

Classic slip: writing $\leq$ at both ends; at $k = 6$ or $k = -6$ the discriminant is zero and the equation has the repeated real root $x = -3$ or $x = 3$, which still counts as a real solution.

50. A solid metal sphere of radius 3 cm is melted down and recast into a solid right circular cylinder of radius 2 cm. What is the height of the cylinder? (Volume of a sphere $= \frac{4}{3}\pi r^3$; volume of a cylinder $= \pi r^2 h$)

- A) 3 cm
- B) 6 cm
- C) 9 cm
- D) 18 cm
- E) 36 cm

Answer: C

Given: sphere of radius 3 cm melted and recast as a cylinder of radius 2 cm; $V(\text{sphere}) = \frac{4}{3}\pi r^3$; $V(\text{cylinder}) = \pi r^2 h$.

Relationship: melting conserves volume, so sphere volume = cylinder volume.

Substitute: $V(\text{sphere}) = \frac{4}{3} \times \pi \times 3^3 = \frac{4}{3} \times 27 \times \pi = 36\pi \text{ cm}^3$

Substitute: $V(\text{cylinder}) = \pi \times 2^2 \times h = 4\pi h$

Equate: $4\pi h = 36\pi$

Result: $h = 36 \div 4 = 9 \text{ cm}$

Classic slip: dividing 36π by 2π instead of 4π , forgetting to square the cylinder's radius, which gives 18 cm.

51. A bag contains 3 red balls and 5 blue balls. Two balls are drawn at random without replacement. What is the probability that the two balls are the same colour?

- A) $3/28$
- B) $5/14$
- C) $13/28$
- D) $15/56$
- E) $15/28$

Answer: C

Given: 3 red and 5 blue balls, 8 in all, two drawn without replacement.

Relationship: $P(\text{both red})$ and $P(\text{both blue})$ are mutually exclusive, so $P(\text{same colour})$ is their sum; each branch multiplies the first-draw and second-draw probabilities.

Substitute red: $(3/8) \times (2/7) = 6/56$

Substitute blue: $(5/8) \times (4/7) = 20/56$

Hop: sum $= 6/56 + 20/56 = 26/56$

Result: $26/56 = 13/28$

Classic slip: computing the complement instead, the $30/56$ chance of two different colours, and reporting $15/28$.

52. A toy train runs along a straight track away from its start point, and its distance from the start is recorded at one-minute intervals: 0 m at 0 min, 120 m at 1 min, 300 m at 2 min and 540 m at 3 min. What is the greatest average speed over any one-minute interval of the journey, expressed in m/s?

- A)** 2.5 m/s **B)** 3.0 m/s **C)** 4.0 m/s **D)** 5.0 m/s **E)** 9.0 m/s

Answer: C

Given: distance from start 0, 120, 300 and 540 m at times 0, 1, 2 and 3 minutes.

Relationship: average speed over one interval = distance gained $\div$ time, with 1 minute = 60 s.

Hop 1: interval 1 covers $120 - 0 = 120$ m, speed = $120 \div 60 = 2.0$ m/s

Hop 2: interval 2 covers $300 - 120 = 180$ m, speed = $180 \div 60 = 3.0$ m/s

Hop 3: interval 3 covers $540 - 300 = 240$ m, speed = $240 \div 60 = 4.0$ m/s

Compare: 4.0 m/s is the largest of the three.

Result: greatest one-minute average speed = 4.0 m/s.

Classic slip: dividing the whole 540 m by the total 180 s to get the overall 3.0 m/s, which blends the intervals instead of isolating the fastest one.

53. What is the value of $\sqrt{50} + \sqrt{18} - \sqrt{8}$, written in the form $a\sqrt{2}$?

- A)** 2 **B)** 4 **C)** 6 **D)** 8 **E)** 10

Answer: C

Given: $\sqrt{50} + \sqrt{18} - \sqrt{8}$, required in the form $a\sqrt{2}$.

Relationship: $\sqrt{a^2 \times b} = a\sqrt{b}$, so extract the largest perfect square from each radicand.

Hop 1: $\sqrt{50} = \sqrt{(25 \times 2)} = 5\sqrt{2}$

Hop 2: $\sqrt{18} = \sqrt{(9 \times 2)} = 3\sqrt{2}$

Hop 3: $\sqrt{8} = \sqrt{(4 \times 2)} = 2\sqrt{2}$

Combine: $(5 + 3 - 2)\sqrt{2} = 6\sqrt{2}$

Result: $a = 6$, value $6\sqrt{2}$

Classic slip: ignoring the minus sign and summing all coefficients to get 10, or flipping the sign of the middle term to get 4.

54. For two events A and B, $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$?

- A)** 0.2 **B)** 0.5 **C)** 0.6 **D)** 0.7 **E)** 0.9

Answer: D

Given: $P(A) = 0.5$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$

Relationship: addition rule, $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$; the subtraction removes the double-counted overlap.

Substitute: $P(A \text{ or } B) = 0.5 + 0.4 - 0.2$

Hop: $0.5 + 0.4 = 0.9$

Result: $0.9 - 0.2 = 0.7$

Classic slip: stopping at 0.9 and never subtracting the 0.2 shared outcomes.

Section E - Physics (Q55-60)

55. A trolley moves along a straight track and its velocity is recorded every 2.0 s: 0 m/s at 0 s, 10 m/s at 2.0 s, 14 m/s at 4.0 s, 14 m/s at 6.0 s and 6 m/s at 8.0 s. What is the magnitude of the trolley's acceleration between 6.0 s and 8.0 s?

- A)** 2 m/s² **B)** 4 m/s² **C)** 5 m/s² **D)** 6 m/s² **E)** 8 m/s²

Answer: B

Given: velocity 14 m/s at 6.0 s; velocity 6 m/s at 8.0 s.

Relationship: $a = \Delta v \div \Delta t$; the magnitude keeps only the size of the answer.

Substitute: $a = (6 - 14) \div (8.0 - 6.0)$

Hop 1: $\Delta v = 6 - 14 = -8 \text{ m/s}$

Hop 2: $\Delta t = 2.0 \text{ s}$, so $a = -8 \div 2.0 = -4 \text{ m/s}^2$

Result: magnitude of acceleration = 4 m/s^2 (the trolley is braking).

Classic slip: quoting the velocity change of 8 m/s as the acceleration without dividing by the 2.0 s interval.

56. A car travels around a circular bend of radius 50 m at a constant speed of 10 m/s. Which of the following statements is correct?

- A)** The acceleration is 2.0 m/s^2 , directed towards the centre of the bend.
B) The acceleration is 2.0 m/s^2 , directed along the direction of travel.
C) The acceleration is 2.0 m/s^2 , directed away from the centre of the bend.
D) The acceleration is zero, because the speed is constant.
E) The acceleration is 0.20 m/s^2 , directed towards the centre of the bend.

Answer: A

Given: radius $r = 50 \text{ m}$; speed $v = 10 \text{ m/s}$; uniform circular motion.

Relationship: centripetal acceleration $a = v^2 \div r$, directed toward the centre at every instant.

Substitute: $a = 10^2 \div 50$

Hop: $10^2 = 100$, so $a = 100 \div 50 = 2.0 \text{ m/s}^2$

Result: acceleration of 2.0 m/s^2 pointing toward the centre of the bend.

Classic slip: dividing without squaring, $10 \div 50 = 0.20 \text{ m/s}^2$, or reasoning that constant speed means zero acceleration although the direction of travel keeps turning.

57. A battery of negligible internal resistance supplies 24 V to an 8.0Ω resistor connected in series with a parallel combination of a 12Ω resistor and a 6.0Ω resistor. What current does the battery supply?

- A)** 1.5 A **B)** 2.0 A **C)** 2.4 A **D)** 3.0 A **E)** 6.0 A

Answer: B

Given: 24 V battery of negligible internal resistance; 8.0Ω in series with a parallel pair of 12Ω and 6.0Ω .

Relationship: parallel pair $R = (R_1 \times R_2) \div (R_1 + R_2)$; series resistances add; $I = V \div R_{\text{total}}$.

Substitute parallel: $(12 \times 6.0) \div (12 + 6.0) = 72 \div 18 = 4.0 \Omega$

Hop: $R_{\text{total}} = 8.0 + 4.0 = 12 \Omega$

Substitute: $I = 24 \div 12$

Result: $I = 2.0 \text{ A}$

Classic slip: dividing 24 V by the lone 8.0Ω resistor to get 3.0 A, forgetting that the whole current must also cross the parallel section.

58. In an insulated container, 0.40 kg of water at 20 °C is mixed with 0.20 kg of water at 65 °C. Assuming no heat is exchanged with the surroundings, what is the final temperature of the mixture?

- A)** 20 °C **B)** 35 °C **C)** 42.5 °C **D)** 50 °C **E)** 65 °C

Answer: B

Given: $m_1 = 0.40$ kg of water at 20 °C; $m_2 = 0.20$ kg of water at 65 °C; insulated container; same specific heat capacity c .

Relationship: heat gained by the cold water = heat lost by the warm water, $m_1c(T - 20) = m_2c(65 - T)$; c cancels.

Substitute: $0.40(T - 20) = 0.20(65 - T)$

Expand: $0.40T - 8.0 = 13.0 - 0.20T$

Collect: $0.40T + 0.20T = 13.0 + 8.0$, so $0.60T = 21.0$

Result: $T = 21.0 \div 0.60 = 35$ °C

Classic slip: taking the plain midpoint $(20 + 65) \div 2 = 42.5$ °C, which weights both samples equally and ignores that the cold one has twice the mass.

59. A farm tank contains a liquid of density 800 kg/m³ filled to a depth of 3.0 m and open to the atmosphere (100 kPa). What is the gauge pressure exerted by the liquid at the bottom of the tank? ($g = 10$ m/s²)

- A)** 2.4 kPa **B)** 8.0 kPa **C)** 24 kPa **D)** 124 kPa **E)** 240 kPa

Answer: C

Given: $\rho = 800$ kg/m³; depth $h = 3.0$ m; $g = 10$ m/s²; atmospheric pressure 100 kPa on the open surface.

Relationship: gauge pressure comes from the liquid column alone, $p = \rho gh$; gauge means the atmosphere is excluded.

Substitute: $p = 800 \times 10 \times 3.0$

Hop: $800 \times 10 = 8000$, so $p = 8000 \times 3.0 = 24000$ Pa

Result: $p = 24$ kPa gauge

Classic slip: adding the 100 kPa of atmospheric pressure, which answers a different question (the absolute pressure, 124 kPa) instead of the gauge pressure asked.

60. An electric kettle rated at 2.0 kW takes 3.0 minutes to bring its water to the boil. How much electrical energy does the kettle transfer in that time?

- A)** 3.6×10^4 J **B)** 3.6×10^5 J **C)** 3.6×10^6 J **D)** 6.0×10^3 J **E)** 6.0×10^5 J

Answer: B

Given: $P = 2.0$ kW; $t = 3.0$ minutes

Relationship: $E = P \times t$, with P in watts and t in seconds.

Convert: 2.0 kW = 2000 W; $t = 3.0 \times 60 = 180$ s

Substitute: $E = 2000 \times 180$

Hop: $2000 \times 180 = 360000$ J

Result: $E = 3.6 \times 10^5$ J

Classic slip: multiplying 2.0 by 3.0 with no unit conversion and stamping the 6.0 as $\times 10^3$ J, mixing kilowatts with minutes.